Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

Show ip route ospf

2

Part 4: Configure OSPF Passive Interfaces

a.

g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? ___ S0/0/0______ 2

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? ___129_____ 2

Does R2 show up as an OSPF neighbor on R1? ____Yes____ 2

Does R2 show up as an OSPF neighbor on R3? ___No_____ 2

What does this information tell you?

____ All traffic destined for 192.168.2.0/24 from R3 will be routed via R1. 2

h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

_ R2(config)# router ospf 1

R2(config-router)# no passive-interface s0/0/1 2

i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? ___S0/0/1___________

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

___65=64+1 64: T1 serial link (1.544Mb/s) 1: R2 Gigabit 0/0 LAN link

____

Is R2 listed as an OSPF neighbor to R3? ____Yes____

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
R1(config-router)# auto-cost reference-bandwidth 100
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

__ To calculate a *more accurate cost metric* for higher-speed links.

____

____

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

192.168.3.0/24: R1 S0/0/1 + R3 G0/0, 781+1=782 192.168.23.0/30: R1 S0/0/1 and R3 S0/0/1, 781+64=845 ____

____

____

i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

___ The total route cost is *1562*.Each serial link has a cost of 781.The route to 192.168.23.0/24 crosses two serial links.Thus, the total cost is computed as:781 × 2 = *1562*

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Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

__

OSPF selects the path with the **lowest total calculated cost.**

The total cost of the path via R2 is lower than that of the direct path from R1 to R3.

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Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

In OSPF, the router ID determines the Designated Router (DR) and Backup Designated Router (BDR) election.

Proper configuration and control of the router ID are critical to ensuring stable and predictable routing behavior, and avoiding unexpected changes caused by interface failures or device reloads.

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2. Why is the DR/BDR election process not a concern in this lab?

The serial links used in this lab are point-to-point links.

DR/BDR election is not required in this lab, as the OSPF topology consists solely of point-to-point links with no multi-access networks that would elect a DR or BDR.

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3. Why would you want to set an OSPF interface to passive?

   This ~~conserves bandwidth by filtering unnecessary~~ OSPF traffic.

   It also prevents the interface from sending OSPF Hello packets and forming adjacencies, while still allowing the connected network to be advertised in OSPF.

   4

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

   _ The use of areas in OSPF enables hierarchical network design, which reduces routing overhead and improves network scalability.

   4

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

   network 10.0.0.0 0.255.255.255 area 0

   4

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

   _ The ~~OSPF route~~ is preferred because it has a lower administrative distance (110) than RIP (120) and EIGRP (internal 90, external 170), making it the selected route.

   EIGRP 0